

Richard Hanna

Woodlyn, PA 19094 | (609) 320-2923 | richardshanna91@gmail.com

Professional Summary

Data science and engineering leader specializing in scalable data products, analytics infrastructure, and automation for complex clinical and research environments. Experienced in technical strategy, application development, ETL architecture, and AI-enabled solutions, with a focus on enabling high-performing teams with building sustainable engineering practices. Open-source software developer and contributor in data analytics and visualization.

Employment History

MANAGER DATA SCIENTIST, 2025 – PRESENT

Children's Hospital of Philadelphia – [CGT Informatics](#), Philadelphia, PA

- Led a team of data scientists and analysts in establishing and hardening development standards and defining programmatic best practices
- Initiated a new work portfolio with new collaborative partnerships in gene therapy supporting data infrastructure and exploratory AI capabilities
- Directed work portfolio, guided prioritization, and defined scope of projects

DATA SCIENTIST II, 2021 – 2025

Children's Hospital of Philadelphia – [CGT Informatics](#), Philadelphia, PA

- Developed novel applications using R Shiny for stem cell transplant reporting
- Engineered databases with error detection and custom analytics for clinical trials
- Authored open-source R packages for data visualization and analytics
- Mentored peers in database engineering and programming best practices

DATA ANALYST & PROGRAMMER, 2019 – 2021

Children's Hospital of Philadelphia – Critical Care Medicine, Philadelphia, PA

- Designed robust ETL processes for international research collaboration
- Developed dynamic dashboards and automated reports using R Shiny which fostered collaboration and led to exploratory analytics in CPR research
- Led initiatives supporting research publications and scientific investigation
- Facilitated data exchange between institutions with varying technical capabilities

RESEARCH PROJECT ENGINEER, 2017 – 2019

Children's Hospital of Philadelphia – Critical Care Medicine, Philadelphia, PA

- Reduced reporting and analytics times using custom programmatic solutions
- Developed a novel method for assessing and modeling pediatric chest geometry
- Built workflows to extract and analyze waveform data from medical devices

PROJECT ENGINEER, 2014 – 2017

GS Medical USA, King of Prussia, PA

- Led R&D lifecycle for several spinal fixation systems
- Ensured ASTM & ISO standard compliance using mechanical testing protocols
- Executed design requests between surgeons and international engineering teams

BIOMECHANICS ENGINEERING RESEARCHER, 2014 – 2015

Children's Hospital of Philadelphia – [CIRP](#), Philadelphia, PA

- Developed a virtual surrogate model for pediatric vehicle restraint systems
- Used motion capture technology to model and analyze occupant crash kinematics

Skills

AWS
Database Design
Dashboard Dev
R
REDCap
Package Dev
Python
SQL

Links

[LinkedIn](#)
[GitHub](#)
[Website](#)
[ORCiD](#)

Certifications

[AWS Solutions Architect](#)
[FE/EIT Certificate](#)

Education

MASTER OF SCIENCE IN BIOMEDICAL ENGINEERING

Drexel University, 2015

GRADUATE CERTIFICATE IN ENGINEERING MANAGEMENT

Drexel University, 2015

BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING

Drexel University, 2015

Selected Publications

Prospective Validation of Multiweek Inpatient Census Forecasting for Pediatric HSCT and Cellular Therapy

[Transplantation](#) and Cellular Therapy

REDCapTidieR: Extracting complex REDCap databases into tidy tables

[The Journal of Open Source Software](#)

A machine-learning model that incorporates CD45 surface expression predicts hematopoietic progenitor cell recovery after freeze–thaw

[Cytotherapy](#)

Automation of Hematopoietic Cell Transplantation Outcomes Reporting Leads to Dramatic Reduction in Errors Reported to Real-World Data Registry

[American Society for Transplantation and Cellular Therapy](#)

Effect of Amplitude Spectral Area on Termination of Fibrillation and Outcomes in Pediatric Cardiac Arrest

[Journal of the American Heart Association](#)

Paediatric In-hospital cardiopulmonary resuscitation quality and outcomes in children with CHD during nights and weekends

[Cardiology in the Young](#)

Risk factors and outcomes for recurrent paediatric in-hospital cardiac arrest: Retrospective multicenter cohort study

[Resuscitation](#)

Related Experience

- **R User Group Steering Committee Member**
Children's Hospital of Philadelphia
- **R/Medicine Steering Committee Member**
The R Consortium
- **ggplot2 Extenders Member**